

POCKETSMART AI

Your Smart Budget & Recommendation Assistant

Project Summary

PocketSmart AI is a GenAI-powered budgeting and recommendation assistant. It helps users plan their spending and receive personalized suggestions within a fixed budget.

1. Abstract

The system combines user budget, preferences and optional images with AI to generate useful recommendations. It focuses on three main areas: Home Interior Planning, Party Planning and Jewelry Planning.

2. Objectives

- Help users manage a fixed budget.
- Generate personalized recommendations.
- Reduce the time needed to search across many platforms.
- Provide simple and user-friendly planning tools.

3. Main Modules

Module	Purpose
Home Planner	Plan home items such as lights, fans, furniture and décor within a budget.
Party Planner	Plan food, decoration, venue and entertainment based on budget and guest count.
Jewelry Planner	Suggest jewelry based on budget, occasion, style and optional outfit image.

4. Technologies Used

- Frontend: HTML, CSS, JavaScript, Jinja2
- Backend: Python with Flask/FastAPI concepts
- AI: Google Gemini API
- APIs: Product and service platforms as required
- Data flow: Frontend → Backend → Gemini AI → Recommendation → Frontend

5. System Architecture

User Input → Frontend → Backend → Gemini AI → Recommendation Logic → Result Display

6. Project Workflow

1. User selects a planner.
2. User enters budget and preferences.
3. Optional image is uploaded where applicable.
4. Backend receives and processes the information.
5. Gemini AI generates recommendations.
6. Results are displayed in a simple format.

7. Functional Requirements

- User registration and login
- Budget input
- Category and preference selection
- Optional image upload
- AI recommendation generation
- Recommendation history

8. Non-Functional Requirements

- Easy-to-use interface
- Fast response
- Secure user sessions
- Mobile-friendly design
- Reliable recommendation processing

9. Advantages

- Saves time
- Supports budget-based decisions
- Combines multiple planning categories
- Provides personalized suggestions
- Simple interface for users

10. Future Enhancements

- Real-time product prices
- More shopping and service APIs
- Voice-based interaction
- Better recommendation history and analytics
- More planning categories

11. Conclusion

PocketSmart AI provides an intelligent way to plan expenses and receive recommendations according to a user's budget and preferences. By combining a web interface, backend services and generative AI, the project demonstrates a practical real-world use of AI for personal budgeting and planning.

Prepared as a simple project reference document